Subar **Mahdi**
Engineering Student |Systems & Mechatronics
linkedin.com/in/smahdi98
subar.mahdi@hotmail.com
My Portfolio

Profile

Engineering student developing multidisciplinary capability across mechanical systems, electronics, embedded systems, robotics and engineering research. Currently contributing to Formula Student vehicle development, competitive rocketry systems engineering, academic research, and an international robotics internship at the University of São Paulo. Former technical operations lead experienced in analysing, optimising and managing complex systems under pressure.

Engineering Experience

Robotics Engineer Intern — University of São Paulo (USP) Jun 2026 – Present

- Contributing to development of a robotic data acquisition platform for Physical AI and autonomous manipulation research
- Supporting integration of instrumented data-glove systems, RGB-D cameras, IMUs and force sensing technologies
- Assisting with ROS 2 workflows, calibration, sensor integration and robotic data collection pipelines
- Working alongside researchers on datasets supporting imitation learning, diffusion policies and Vision-Language-Action models
- Research activity includes potential IEEE publication outcomes

Mechanical and Electrical Team Member — Bristol Racing (Formula Student) Sep 2025 – Jul 2026

- Working within the mechanical and electrical team on a scaled Formula Student RC vehicle
- Supporting drivetrain packaging, steering concepts, CAD assemblies and wiring layouts
- Assisting design-for-manufacture using laser-cut and 3D-printed components
- Developing understanding of reliability, tolerances, testing and performance trade-offs

Systems Engineer (Feed Systems) — HyPower Rocket Society Sep 2025 – Jul 2026

- Supporting development of pressurisation and propellant feed systems for competitive rocketry
- Producing and interpreting P&IDs and CAD models for tanks, valves and feed routing
- Investigating pressure-drop behaviour and subsystem integration considerations
- Contributing to safety-driven design reviews and systems-level engineering decisions

Junior Researcher — Academic Research Society (Engineering & Technology) Sep 2025 – Jul 2026

- Conducting applied engineering and technology research
- Producing literature reviews, technical summaries and structured research outputs
- Currently investigating self-healing composite materials for aerospace structural applications

Projects

Smart Thermal Temperature Monitoring System

Designed and built an Arduino-based thermal monitoring system integrating a DS18B20 temperature sensor, AMG8833 thermal camera array, LCD display and threshold-based alert logic. Conducted experimental testing, analysed sensor performance, compared point sensing against spatial thermal imaging and produced a full technical dissertation and research poster.

Smart Lamp — Embedded Electromechanical System

Designed and developed an Arduino-controlled smart lamp incorporating sensors, LED control, MOSFET switching and embedded firmware. Developed practical skills in hardware debugging, wiring, power delivery, enclosure design and embedded systems integration.

Load-Bearing Single-Axis Linear Actuator (Ongoing)

Designing a mechanical actuator system to develop understanding of structural loading, motion transmission, stiffness, tolerances, backlash and real-world mechanical reliability. Producing CAD assemblies and evaluating design trade-offs throughout development.

Professional Experience

Programmatic Advertising Lead — Global Media Organisations May 2021 – Oct 2025

- Led delivery and optimisation of large-scale technical advertising systems across multiple global platforms
- Analysed performance data, diagnosed system-level issues and implemented optimisation strategies
- Coordinated technical and commercial stakeholders under demanding operational deadlines
- Managed complex workflows requiring structured problem solving, analytical reasoning and systems thinking

- Delivered measurable improvements in efficiency, performance and return on investment

Technical Skills

- **CAD & Mechanical Design :** Fusion 360, Autodesk Inventor, SolidWorks (basic), Mechanical Design, Design for Manufacture
- **Electronics & Embedded Systems :** Arduino, Embedded C/C++, Sensor Integration, PWM Control, MOSFET Switching, Thermal Sensors
- **Programming & Software :** Python, MATLAB, GitHub, ROS 2 (foundational)
- **Manufacturing :** FDM 3D Printing, Laser Cutting, Prototype Assembly
- **Engineering Analysis :** Experimental Testing, Data Analysis, Technical Reporting, Research Methods

Affiliations :

IMechE Student Affiliate • Bristol Racing • HyPower Rocket Society • Academic Research Society • Engineers Without Borders

Education

CertHE Foundation in Science, Engineering and Mathematics - University of Bristol — Sept 2025 - June 2026

Digital Marketing Bootcamp — Multiverse

Codecademy — Python 2 & Python 3